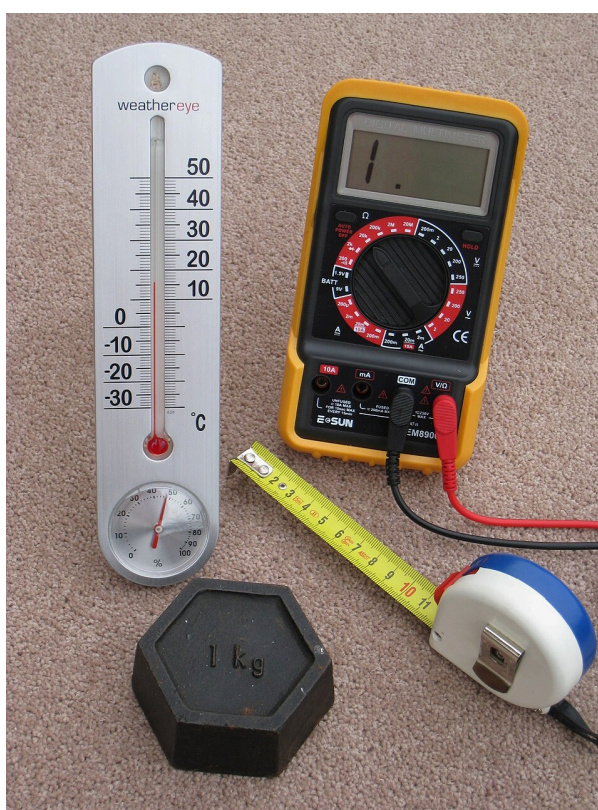


Метрическая система

Система измерения — это система, в которой физические величины (такие как длина, вес, температура и т. д.) могут быть выражены в количественном отношении с помощью чисел. Среди множества систем измерения **метрическая система** является одной из самых распространённых. В ней **стандартизирован** набор основных единиц измерения и **номенклатура**, описывающая относительно большие и малые величины с помощью **десятичных** мультипликативных **префиксов** (таких как *кило* и *милли*). **Международная система единиц** (СИ) — хороший пример, но есть и другие широко распространённые системы (менее популярные, чем СИ), которые используются в основном в конкретных областях, например **гауссовы единицы** для **электромагнетизма**.



Масса **килограмма** и три метрических измерительных прибора: **рулетка** в **сантиметрах**, **термометр** в **градусах Цельсия** и **мультиметр**, который измеряет потенциал в **вольтах**, силу тока в **амперах** и сопротивление в **омах**.

Хотя правила, регулирующие метрическую систему, со временем менялись, современное определение Международной системы единиц предписывает использовать **метрические приставки** и семь основных единиц: **метр** (м), **килограмм** (кг), **секунда** (с), **ампер** (А), **кельвин** (К), **моль** (моль) и **кандела** (кд).^[1]

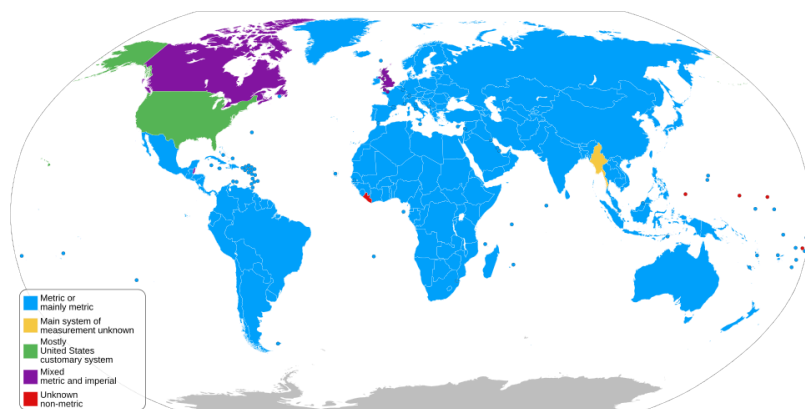
Производные единицы СИ — это именованные комбинации основных единиц, такие как **герц** (количество циклов в секунду), **ньютон** ($\text{кг}\cdot\text{м}/\text{с}^2$) и **тесла** ($1\text{ кг}\cdot\text{с}^{-2}\cdot\text{А}^{-1}$). В случае с

градусами **Цельсия** (°C) это сдвинутая шкала, производная от шкалы Кельвина (273,15 К равно 0 °C, а интервал в 1 К равен интервалу в 1 °C).

Система СИ произошла от более старой системы единиц **метр-килограмм-секунда** (МКС), хотя определения основных единиц со временем менялись. Сегодня все основные единицы определяются **физическими константами**, а не прототипами в виде физических объектов, как это было раньше.

К другим вариантам метрической системы относятся **система единиц сантиметр-грамм-секунда**, **система единиц метр-тонна-секунда** и **гравитационная метрическая система**. В каждой из них есть **не входящие в СИ метрические единицы**, и некоторые из этих систем до сих пор используются в ограниченном контексте. Отдельные единицы измерения, не входящие в СИ, такие как **литр**, по-прежнему широко распространены.^[2]

Принятие



Единицы измерения, используемые в повседневной жизни, по странам на 2019 год

Система СИ принята в качестве официальной системы мер и весов почти во всех странах мира.

Заметным исключением являются Соединенные Штаты Америки (США). Несмотря на то, что в некоторых случаях они используют эту систему, США сопротивляются ее повсеместному внедрению и продолжают использовать **другие системы измерения**.^[3]

Adopting the metric system is known as **metrication**.

Мультипликативные префиксы

В системе СИ и, как правило, в старых метрических системах кратные и дольные единицы измерения могут обозначаться префиксом в названии единицы измерения, который указывает на десятичный (основание 10) множитель.

The prefix *kilo*, for example, implies a factor of 1000 (10^3), and the prefix *milli* implies a factor of 1/1000 (10^{-3}). Thus, a *kilometre* is a thousand metres, and a *milligram* is one thousandth of a gram. These relations can be written symbolically as:^[4]

$$1\text{ km} = 1000\text{ m}$$
$$1\text{ mg} = 0.001\text{ g}$$

Основные единицы измерения

Десятичная система основана на метре, который был введён во Франции в 1790-х годах. Кульминацией исторического развития этих систем стало определение Международной системы единиц (СИ) в середине XX века под контролем международного органа по стандартизации.

Историческая эволюция метрических систем привела к признанию нескольких принципов. Выбирается набор независимых физических величин, в терминах которых можно выразить все природные величины, называемые базовыми. Для каждой из этих величин определяется репрезентативная величина в качестве базовой единицы измерения. Определение базовых единиц все чаще реализуется на основе фундаментальных природных явлений, а не копий физических артефактов. Единица измерения, выведенная на основе основных единиц, используется для выражения величин, размерность которых может быть выведена из основных размерностей системы. Например, квадратный метр — это производная единица измерения площади, которая выводится из длины. Эти производные единицы измерения согласованны, то есть в них используются только произведения степеней основных единиц измерения без каких-либо дополнительных множителей. Для любой величины, единица измерения которой имеет название и обозначение, определяется расширенный набор единиц измерения меньшего и большего размера, связанных между собой коэффициентами в степенях десяти. Единицей времени должна быть секунда; единицей длины должен быть метр или его десятичная кратная величина; единицей массы должен быть грамм или его десятичная кратная величина.

Metric systems have evolved since the 1790s, as science and technology have evolved, in providing a single universal measuring system. Before and in addition to the SI, other metric

Метрические приставки в повседневном использовании

Префикс	Символ	Фактор	Сила
тера	T	1 000 000 000 000	10^{12}
гига	G	1 000 000 000	10^9
мега	M	1 000 000	10^6
kilo	k	1 000	10^3
hecto	h	100	10^2
deca	da	10	10^1
(none)	(none)	1	10^0
deci	d	0.1	10^{-1}
centi	c	0.01	10^{-2}
milli	m	0.001	10^{-3}
micro	μ	0.000 001	10^{-6}
nano	n	0.000 000 001	10^{-9}
pico	p	0.000 000 000 001	10^{-12}

systems include: the [MKS system of units](#) and the [MKSA](#) systems, which are the direct forerunners of the SI; the [centimetre–gram–second \(CGS\)](#) system and its subtypes, the [CGS electrostatic](#) (cgs-esu) system, the [CGS electromagnetic](#) (cgs-emu) system, and their still-popular blend, the [Gaussian system](#); the [metre–tonne–second \(MTS\)](#) system; and the [gravitational metric systems](#), which can be based on either the metre or the centimetre, and either the gram, gram-force, kilogram or kilogram-force.

Атрибуты

Простота освоения и использования

Метрическая система призвана быть простой в использовании и широко применимой. Она включает в себя единицы измерения, основанные на природных явлениях, десятичные соотношения, приставки для обозначения кратных и дольных единиц, а также структуру основных и производных единиц.

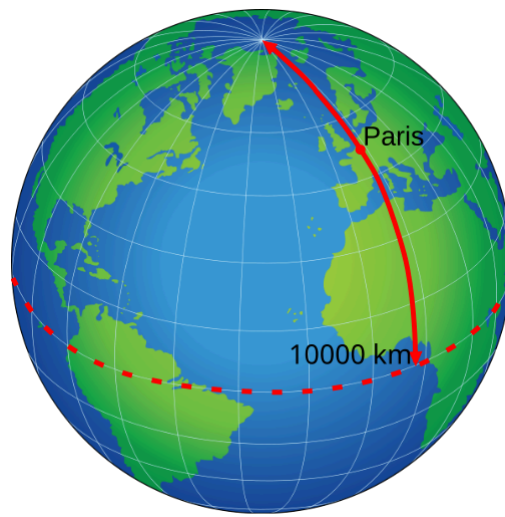
It is a [coherent system](#) with [derived units](#) built from base units using logical rather than empirical relationships and with multiples and submultiples of both units based on decimal factors and identified by a [common set of prefixes](#).^{[5]:15–18}

Extensibility

The metric system is extensible since the governing body reviews, modifies and extends it needs arise. For example, the [katal](#), a derived unit for catalytic activity equivalent to one [mole](#) per second (1 mol/s), was added in 1999.^[6]

Realisation

The base units used in a measurement system must be [realisable](#). To that end, the definition of each SI base unit is accompanied by a *mise en pratique* (practical realisation) that describes at least one way that the unit can be measured.^[7] Where possible, definitions of the base units were developed so that any laboratory equipped with proper instruments would be able to realise a standard without reliance on an artefact held by another country. In practice, such realisation is done under the auspices of a [mutual acceptance arrangement](#).^[8]



The **metre** was originally defined to be one *ten millionth* of the distance between the **North Pole** and the **Equator** through **Paris**.^[9]

In 1791 the commission originally defined the **metre** based on the size of the earth, equal to one ten-millionth of the distance from the equator to the North Pole. In the SI, the standard metre is now defined as exactly $\frac{1}{299,792,458}$ of the distance that light travels in a **second**.^{[10][11]} The metre can be realised by measuring the length that a light wave travels in a given time, or equivalently by measuring the wavelength of light of a known frequency.^[12]

The **kilogram** was originally defined as the mass of one cubic decimetre of water at 4 °C, standardised as the mass of a man-made artefact of platinum–iridium held in a laboratory in France, which was used until a **new definition was introduced in May 2019**. Replicas made in 1879 at the time of the artefact's fabrication and distributed to signatories of the **Metre Convention** serve as *de facto* standards of mass in those countries. Additional replicas have been fabricated since as additional countries have joined the convention. The replicas were subject to periodic validation by comparison to the original, called the **IPK**. It became apparent that either the IPK or the replicas or both were deteriorating, and are no longer comparable: they had diverged by 50 µg since fabrication, so figuratively, the accuracy of the kilogram was no better than 5 parts in a hundred million or a relative accuracy of 5×10^{-8} . The **revision of the SI** replaced the IPK with an exact definition of the **Planck constant** as expressed in SI units, which defines the kilogram in terms of fundamental constants.^{[13][14][15]}

Base and derived unit structure

A base quantity is one of a conventionally chosen subset of physical quantities, where no quantity in the subset can be expressed in terms of the others. A base unit is a unit adopted for expressing a base quantity. A derived unit is used for expressing any other quantity, and is a product of powers of base units. For example, in the modern metric system, length has the unit metre and time has the unit second, and speed has the derived unit metre per second.^{[5]:15} Density, or mass per unit volume, has the unit kilogram per cubic metre.^{[5]:434}

Decimal ratios

A significant characteristic of the metric system is its use of decimal multiples – powers of 10. For example, a length that is significantly longer or shorter than 1 metre can be represented in units that are a power of 10 or 1000 metres. This differs from many older systems in which the ratio of different units varied. For example, 12 [inches](#) is one [foot](#), but the larger unit in the same system, the [mile](#) is not a power of 12 feet. It is 5,280 feet – which is hard to remember for many.^{[5]:17}

In the early days, multipliers that were positive powers of ten were given Greek-derived prefixes such as *kilo-* and *mega-*, and those that were negative powers of ten were given Latin-derived prefixes such as *centi-* and *milli-*. However, 1935 extensions to the prefix system did not follow this convention: the prefixes *nano-* and *micro-*, for example have Greek roots.^{[16]:222–223} During the 19th century the prefix *myria-*, derived from the Greek word *μύριοι* (*mýrioi*), was used as a multiplier for 10 000.^[17]

When applying prefixes to derived units of area and volume that are expressed in terms of units of length squared or cubed, the square and cube operators are applied to the unit of length including the prefix, as illustrated below.^[4]

$$1 \text{ mm}^2 \text{ (square millimetre)} = (1 \text{ mm})^2 = (0.001 \text{ m})^2 = 0.000\,001 \text{ m}^2$$

$$1 \text{ km}^2 \text{ (square kilometre)} = (1 \text{ km})^2 = (1000 \text{ m})^2 = 1\,000\,000 \text{ m}^2$$

$$1 \text{ mm}^3 \text{ (cubic millimetre)} = (1 \text{ mm})^3 = (0.001 \text{ m})^3 = 0.000\,000\,001 \text{ m}^3$$

$$1 \text{ km}^3 \text{ (cubic kilometre)} = (1 \text{ km})^3 = (1000 \text{ m})^3 = 1\,000\,000\,000 \text{ m}^3$$

For the most part, the metric prefixes are used uniformly for SI units, but for large measures of seconds, the non-SI units of [minute](#), [hour](#) and [day](#) are customary instead.^[2]

Coherence



James Clerk Maxwell.

James Clerk Maxwell played a major role in developing the concept of a coherent CGS system and in extending the metric system to include electrical units.

Each variant of the metric system has a degree of coherence—the derived units are directly related to the base units without the need for intermediate conversion factors.^[18] For example, in a coherent system the units of **force**, **energy**, and **power** are chosen so that the equations

$$\text{force} = \text{mass} \times \text{acceleration}$$

$$\text{energy} = \text{force} \times \text{distance}$$

$$\text{energy} = \text{power} \times \text{time}$$

hold without the introduction of unit conversion factors. Once a set of coherent units has been defined, other relationships in physics that use this set of units will automatically be true.

Therefore, Einstein's mass–energy equation, $E = mc^2$, does not require extraneous constants when expressed in coherent units.^[19]

The CGS system had two units of energy, the **erg** that was related to **mechanics** and the **calorie** that was related to **thermal energy**; so only one of them (the erg) could bear a coherent relationship to the base units. Coherence was a design aim of SI, which resulted in only one unit of energy being defined – the **joule**.^[20]

Rationalisation

Maxwell's equations of electromagnetism contained a factor of $1/(4\pi)$ relating to **steradians**, representative of the fact that electric charges and magnetic fields may be considered to emanate from a point and propagate equally in all directions, i.e. spherically. This factor made equations more awkward than necessary, and so Oliver Heaviside suggested adjusting the system of units to remove it.^[21]

Повседневные понятия

Основные единицы метрической системы всегда отражали распространенные в природе величины или соотношения, даже с учетом современных уточнений в определениях и методологии. В случаях, когда лабораторная точность не требуется или недоступна, а также когда достаточно приблизительных значений, можно обойтись общепринятыми понятиями.

Time, length, volume and mass

The second is readily determined from the Earth's rotation period. Unlike other units, time multiples are not decimal. A second is $\frac{1}{60}$ of a minute, which is $\frac{1}{60}$ of an hour, which is $\frac{1}{24}$ of a day, so a second is $\frac{1}{86\,400}$ of a day.

The length of the [equator](#) is close to 40 000 000 m (more precisely 40 075 014.2 m).^[22] In fact, the dimensions of our planet were used by the French Academy in the original definition of the metre.^[23] A dining tabletop is typically about 0.75 metres high.^[24] A very tall human is about 2 metres tall.^[25]

The mass of a litre of cold water is very close to 1 kilogram.^[26] Therefore, 1 millilitre of water occupies 1 cubic centimetre and has a mass of 1 gram.

A [1-euro coin](#) has a mass of 7.5 g;^[27] a [Sacagawea US 1-dollar coin](#) has a mass of 8.1 g;^[28] the mass of a [UK 50-pence coin](#) is 8.0 g.^[29]

Temperature

In everyday use, the Celsius scale is more commonly used than the Kelvin scale; however, a temperature difference of one kelvin is the same as one degree Celsius. A temperature in kelvins is the temperature on the Celsius scale plus about 273. Water freezes at about 0 °C (273 K) and boils at about 100 °C (373 K) at sea level; [human body temperature](#) is about 37 °C (310 K).

Brightness

A candela is about the luminous intensity of a moderately bright candle, or 1 candle power. A 60 [watt tungsten-filament incandescent light bulb](#) has a luminous intensity of about 800 lumens^[30] which is radiated equally in all directions (i.e. 4π steradians), thus is equal to

$$I_v = \frac{800 \text{ lm}}{4\pi \text{ sr}} \approx 64 \text{ cd}.$$

Electricity

A 60 W incandescent light bulb consumes 0.5 A at 120 V (US mains voltage). A 60 W bulb rated at 230 V (European mains voltage) consumes 0.26 A at this voltage. This is evident from the formula $P = I V$.

Mole and mass relationship

A mole of a substance has a mass that is its [molecular mass](#) expressed in units of grams. The mass of a mole of carbon is 12.0 g, and the mass of a mole of table salt is 58.4 g.

Since all gases have the same volume per mole at a given temperature and pressure far from their points of liquefaction and solidification (see [Perfect gas](#)), and air is about $\frac{1}{5}$ oxygen (molecular mass 32) and $\frac{4}{5}$ nitrogen (molecular mass 28), the density of any near-perfect gas relative to air can be obtained to a good approximation by dividing its molecular mass by 29 (because $\frac{4}{5} \times 28 + \frac{1}{5} \times 32 = 28.8 \approx 29$). For example, [carbon monoxide](#) (molecular mass 28) has almost the same density as air.

История

Метрическая система стала результатом многочисленных попыток стандартизации мер и весов, предпринимавшихся на протяжении многих веков, поскольку более ранние (китайская, египетская, греческая, римская и другие) системы мер и весов взаимодействовали друг с другом в рамках мировой торговли.

17th & 18th centuries

Around 1665–1670, Gabriel Mouton, a priest of the Collegiate Church of St Paul in [Lyon](#), France, suggested that the many different systems of units then in use in France should be replaced by a single all-embracing decimal system. He proposed that linear measurements would be based on a natural standard of length, namely the length of one minute of arc of the largest circle that could be drawn round the Earth, and which he proposed to call one milliarc. Mouton suggested that the unit of mass should be based on the unit of length, and on the mass density of water. While Mouton's contribution is sometimes overlooked, these ideas were clearly the basis over 120 years later, when the metric system was devised and standardised in France.^{[31]:140–142} During the final days of France's *Ancien régime* (1775-1788) and the early days of the [French Revolution](#) (1789–99), the first metric system was devised by chemists [Antoine](#) and [Anne-Marie Lavoisier](#) for the benefit of the [Ferme générale](#) in return for financial support of Antoine Lavoisier's research.

Following the model established by Mouton, Lavoisier's unit of length, the metre, was based on the dimensions of the [Earth](#), and the unit of [mass](#), the kilogram, was based on the mass of 1 [litre](#) of water (the [volume](#) of water which occupies 1 cubic [decimetre](#) of space, using the same unit of length).^{[31]:140–142}



[Pavillon de Breteuil](#), Saint-Cloud, France, the home of the metric system's *Mètre des Archives* and *Kilogramme des Archives* since 1875

The backwardness of the Ancien Régime left France with a need to reform its numerous systems of various local weights and measures. In 1790, [Charles Maurice de Talleyrand-Périgord](#) proposed Lavoisier's system based on natural units to the [French National Assembly](#), without reference to the prior work of Mouton and Lavoisier. The aim was global adoption of the metric system. With the [United Kingdom](#) not responding to requests to collaborate in the development of the system, the [French Academy of Sciences](#) established a commission to implement this new standard alone, and in 1799, the new system was launched in France.^{[31]:145–149}

A number of different metric systems have since been developed, all using the *Mètre des Archives* and *Kilogramme des Archives* (or their descendants) as their base units, but differing in the definitions of the various derived units.

19th century

In 1832, [Gauss](#) used the astronomical second as a base unit in defining the gravitation of the Earth, and together with the milligram and millimetre, this became [the first system of mechanical units](#). He showed that the strength of a magnet could also be quantified in terms of these units, by measuring the oscillations of a magnetised needle and finding the quantity of "magnetic fluid" that produces an acceleration of one unit when applied to a unit mass.^{[32][33]} The [centimetre–gram–second system of units](#) (CGS) was the first coherent metric system, having been developed in the 1860s and promoted by [James Clerk Maxwell](#) and [Lord Kelvin](#). In 1874, this system was formally promoted by the [British Association for the Advancement of Science](#) (BAAS).^[34] The system's characteristics are that density is expressed in g/cm^3 , force expressed in [dynes](#) and mechanical energy in [ergs](#). Thermal energy was defined in [calories](#), one calorie being the energy required to raise the temperature of one gram of water from 15.5 °C to 16.5 °C.

The Association also recognised [two sets of units for electrical and magnetic properties](#) – the electrostatic set of units and the electromagnetic set of units.^[35]

The CGS units of electricity were cumbersome to work with. This was remedied at the 1893 International Electrical Congress held in Chicago by defining the "international" ampere and ohm using definitions based on the [metre](#), [kilogram](#) and [second](#), in the [International System of Electrical and Magnetic Units](#).^[36] During the same period in which the CGS system was being extended to include electromagnetism, other systems were developed, distinguished by their choice of coherent base unit, including the [Practical System of Electric Units](#), or QES (quad–eleventhgram–second) system. Here, the base units are the quad, equal to 10^7 m (approximately a quadrant of the Earth's circumference), the eleventhgram, equal to 10^{-11} g, and the second. These were chosen so that the corresponding electrical units of potential difference, current and resistance had a convenient magnitude.^{[37]:268[38]:17}

Variants of the metric system

Measure	SI/MKS	CGS	MTS
distance	metre (m)	centimetre (cm)	metre (m)
mass	kilogram (kg)	gram (g)	tonne (t)
time	second (s)	second (s)	second (s)
velocity	m/s	cm/s	m/s
acceleration	m/s ²	gal (Gal)	m/s ²
force	newton (N)	dyne (dyn)	sthene (sn)
pressure	pascal (Pa)	barye (Ba)	pièze (pz)
energy	joule (J)	erg (erg)	kilojoule (kJ)
power	watt (W)	erg/s (erg/s)	kilowatt (kW)
viscosity	Pa·s	poise (P)	pz·s

20th century

In 1901, [Giovanni Giorgi](#) showed that by adding an electrical unit as a fourth base unit, the various anomalies in electromagnetic systems could be resolved. The metre–kilogram–second–[coulomb](#) (MKSC) and metre–kilogram–second–[ampere](#) (MKSA) systems are examples of such systems.^{[39][21]}

The [metre–tonne–second system of units](#) (MTS) was based on the metre, [tonne](#) and second – the unit of force was the [sthène](#) and the unit of pressure was the [pièze](#). It was invented in France for industrial use and from 1933 to 1955 was used both in France and in the [Soviet Union](#).^{[40][41]} [Gravitational metric systems](#) use the [kilogram-force](#) (kilopond) as a base unit of force, with mass measured in a unit known as the [hyl](#), *Technische Masseneinheit* (TME), mug or [metric slug](#).^[42] Although the CGPM passed a resolution in 1901 defining the standard value of [acceleration due to gravity](#) to be 980.665 cm/s^2 , gravitational units are not part of the [International System of Units](#) (SI).^[43]

Current

The [International System of Units](#) is the modern metric system. It is based on the metre–kilogram–second–ampere (MKSA) system of units from early in the 20th century.^[20] It also includes numerous coherent derived units for common quantities like power (watt) and irradiance (lumen). Electrical units were taken from the International system then in use. Other units like those for energy (joule) were modelled on those from the older CGS system, but scaled to be coherent with MKSA units. Two additional base units – the *kelvin*, which is equivalent to degree Celsius for change in thermodynamic temperature but set so that 0 K is [absolute zero](#), and the *candela*, which is roughly equivalent to the [international candle](#) unit of illumination – were introduced. Later, another base unit, the *mole*, a unit of [amount of substance](#) equivalent to the [Avogadro number](#) number of specified molecules, was added along with several other derived units.^[44]

The system was promulgated by the General Conference on Weights and Measures (French: *Conférence générale des poids et mesures* – CGPM) in 1960. At that time, the metre was redefined in terms of the wavelength of a spectral line of the [krypton-86](#) atom (krypton-86 being a stable isotope of an inert gas that occurs in undetectable or trace amounts naturally), and the standard metre artefact from 1889 was retired.^{[5]:16}

Today, the International System of Units consists of 7 base units and innumerable coherent derived units including 22 with special names. The last new derived unit, the *katal* for catalytic activity, was added in 1999. All the base units except the second are now defined in terms of exact and invariant constants of physics or mathematics, barring those parts of their definitions which are dependent on the second itself. As a consequence, the speed of light has now become an exactly defined constant, and defines the metre as $1/299,792,458$ of the distance light travels in a second. The kilogram was defined by a [cylinder of platinum-iridium alloy](#) until a [new definition](#) in terms of [natural physical constants](#) was adopted in 2019. As of 2022, the range of decimal prefixes has been extended to those for 10^{30} (*quetta*–) and 10^{-30} (*quecto*–).^[45]

См. также

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- [Binary prefix](#) – Prefix indicating a power of two
 - [Electrostatic units](#) – Variant of the metric system
 - [History of measurement](#)
 - [ISO 31](#) – Superseded standard on quantities and units
 - [ISO/IEC 80000](#) – International standard on physical quantities and units of measurement
 - [List of metric units](#) – Class of units of measurement
 - [Metrology](#) – Science of measurement and its application

- [Outline of metrology and measurement](#) – Topical index of English Wikipedia articles about metrology and measurement
- [Preferred metric sizes](#) – Metricated industry standards
- [Unified Code for Units of Measure](#) – System of codes for unambiguously representing measurement units

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